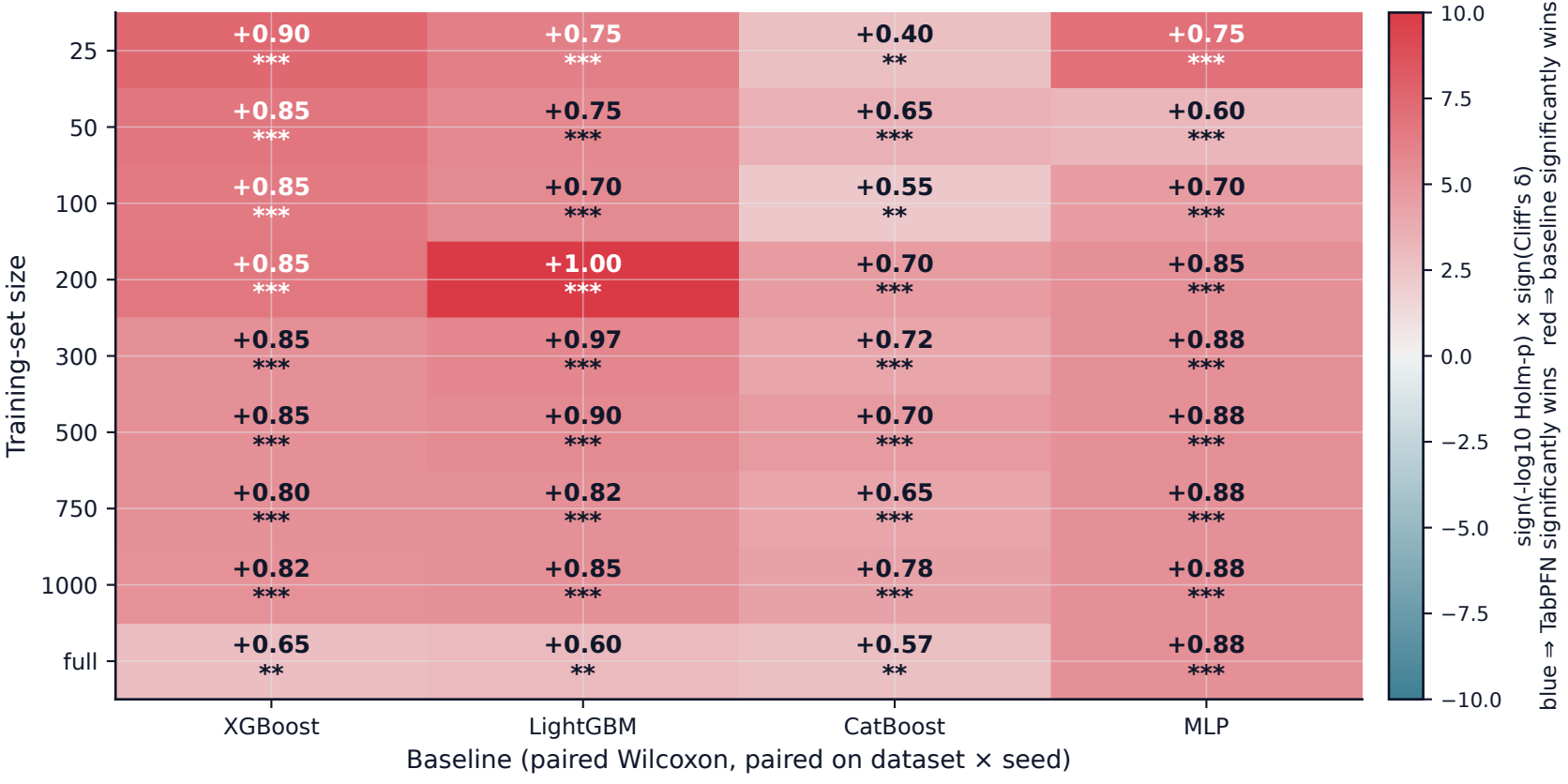

Statistical significance of TabPFN advantage per N
(*** p<0.001 · ** p<0.01 · * p<0.05 · ns p≥0.05 after Holm correction)
Training-set size
25
50
100
200
300
500
750
1000
full
+0.90 *** +0.75 *** +0.40 ** +0.75 ***
+0.85 *** +0.75 *** +0.65 *** +0.60 ***
+0.85 *** +0.70 *** +0.55 ** +0.70 ***
+0.85 *** +1.00 *** +0.70 *** +0.85 ***
+0.85 *** +0.97 *** +0.72 *** +0.88 ***
+0.85 *** +0.90 *** +0.70 *** +0.88 ***
+0.80 *** +0.82 *** +0.65 *** +0.88 ***
+0.82 *** +0.85 *** +0.78 *** +0.88 ***
+0.65 ** +0.60 ** +0.57 ** +0.88 ***
XGBoost
LightGBM
CatBoost
MLP
Baseline (paired Wilcoxon, paired on dataset × seed)
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
sign(-log10 Holm-p) × sign(Cliff's δ)
blue ⇒ TabPFN significantly wins red ⇒ baseline significantly wins